lxNavbar
lx-
Header
lx-
Footer

Hyperconvexity and domains of discontinuity for surface groups in $SL_3\mathbb{C}$

BY

Jennifer Vaccaro
BS, Olin College of Engineering, 2017
MS, University of Illinois at Chicago, 2022

THESIS [PhD candidates may use DISSERTATION here instead]

Submitted as partial fulfillment of the requirements
for the degree of Doctorate of Philosophy in Mathematics
in the Graduate College of the
University of Illinois Chicago, 2026

Chicago, Illinois

Defense Committee:
Emily Dumas
Name
Name
Name, Outside Members Need Affiliation Here

Accessibility Statement

An accessible EPUB version of this document is available at [URL] or can be obtained by contacting [contact details].

Dedication

For my grandmother Martha, who also saw the joy in geometry.

Acknowledgements

Emily (integrity, knowledge, high standards)
Sam (love, support, meals, no pressure)
Family (you don't know why I wanted to do this, but that's ok)
Friends (kept me company, kept me happy)
Other math teachers and mentors
Gertrude (tiniest companion)

Preface

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Curabitur pretium tincidunt lacus. Nulla gravida orci a odio. Nullam varius, turpis et commodo pharetra, est eros bibendum elit, nec luctus magna felis sollicitudin mauris.

Integer in mauris eu nibh euismod gravida. Duis ac tellus et risus vulputate vehicula. Donec lobortis risus a elit. Etiam tempor. Ut ullamcorper, ligula eu tempor congue, eros est euismod turpis, id tincidunt sapien risus a quam.

Contents

Accessibility Statement ii

Dedication iii

Acknowledgements iv

Preface vi

List of Figures x

List of Tables xi

List of Abbreviations xii

Summary xiv

Chapter 1. Introduction 1
1. Motivation 2

Chapter 2. Background 4
1. Group theory 5

Chapter 3. Main Result 6
1. Richardson varieties 7
2. Assumptions and notation 8
3. Statement of Theorem 3.1 and related results 8
4. Proof of Theorem 3.1. 8
5. Action of $\rho(g)$ on F^3 9
6. F preserves Richardson 0- and 1-varieties of $\xi(g^+), \xi(g^-)$ 10
7. F preserves Richardson 0-varieties of Λ 10
8. F preserves Richardson 1-varieties of Λ 11
9. Comments 12

Chapter 4. Another result 13
1. Notation 14
2. Statement of Theorem 4.1 and related results 14
3. Key ingredients 14
4. Proof of Theorem 4.1 15
5. Topological conjugacies of reducible representations 15

Chapter 5. Vitae 16

Appendix A. Technical Lemmas 18

Bibliography 20

List of Figures

1 This is a torus 2
2 The Snake Lemma 2

List of Tables

1 Example Table 3

List of Abbreviations

- $\mathbb{R}$ The real numbers
- $\mathbb{C}$ The complex numbers

Summary

Cras mollis scelerisque nunc. Nullam arcu. Aliquam consequat. Curabitur augue lorem, dapibus quis, laoreet et, pretium ac, nisi. Aenean magna nisl, mollis quis, molestie eu, feugiat in, orci. In hac habitasse platea dictumst.

Maecenas fermentum consequat mi. Donec fermentum. Pellentesque malesuada nulla a mi. Duis sapien sem, aliquet nec, commodo eget, consequat quis, neque. Aliquam faucibus, elit ut dictum aliquet, felis nisl adipiscing sapien, sed malesuada diam lacus eget erat.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Vivamus luctus urna sed urna ultricies ac tempor dui sagittis. In condimentum facilisis porta.

Sed nec diam eu diam mattis viverra. Nulla fringilla, orci ac euismod semper, magna diam porttitor mauris, quis sollicitudin sapien justo in libero.

Fusce lacinia quam nisi, euismod ultrices nulla malesuada ut. Curabitur sit amet magna quam. Praesent in libero vel turpis pellentesque egestas sit amet vel nunc.

Morbi lectus risus, iaculis vel, suscipit quis, luctus non, massa. Fusce ac turpis quis ligula lacinia aliquet. Mauris ipsum. Nulla metus metus, ullamcorper vel, tincidunt sed, euismod in, nibh.

Quisque volutpat condimentum velit. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos. Nam nec ante. Sed lacinia, urna non tincidunt mattis, tortor neque adipiscing diam, a cursus ipsum ante quis turpis.

Cras elementum ultrices diam. Maecenas ligula massa, varius a, semper congue, euismod non, mi. Proin porttitor, orci nec nonummy molestie, enim est eleifend mi, non fermentum diam nisl sit amet erat.

Duis semper. Duis arcu massa, scelerisque vitae, consequat in, pretium a, enim. Pellentesque congue. Ut in risus volutpat libero pharetra tempor. Cras vestibulum bibendum augue.

Praesent egestas leo in pede. Praesent blandit odio eu enim. Pellentesque sed dui ut augue blandit sodales. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae.

Aliquam nibh. Mauris ac mauris sed pede pellentesque fermentum. Maecenas adipiscing ante non diam sodales hendrerit.

Ut velit mauris, egestas sed, gravida nec, ornare ut, mi. Aenean ut orci vel massa suscipit pulvinar. Nulla sollicitudin. Fusce varius, ligula non tempus aliquam, nunc turpis ullamcorper nibh, in tempus sapien eros vitae ligula.

Pellentesque rhoncus nunc et augue. Integer id felis. Curabitur aliquet pellentesque diam. Integer quis metus vitae elit lobortis egestas.

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Morbi vel erat non mauris convallis vehicula. Nulla et sapien. Integer tortor tellus, aliquam faucibus, convallis id, congue eu, quam.

Mauris ullamcorper felis vitae erat. Proin feugiat, augue non elementum posuere, metus purus iaculis lectus, et tristique ligula justo vitae magna.

Aliquam convallis sollicitudin purus. Praesent aliquam, enim at fermentum mollis, ligula massa adipiscing nisl, ac euismod nibh nisl eu lectus.

Fusce vulputate sem at sapien. Vivamus leo. Aliquam euismod libero eu enim. Nulla nec felis sed leo placerat imperdiet. Aenean suscipit nulla in justo.

Suspendisse cursus rutrum augue. Nulla tincidunt tincidunt mi. Curabitur iaculis, lorem vel rhoncus faucibus, felis magna fermentum augue, et ultricies lacus lorem varius purus.

Sed adipiscing ornare risus. Morbi est est, blandit sit amet, sagittis vel, euismod vel, velit. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

Nullam nonummy metus. Vestibulum volutpat pretium libero. Cras id dui. Aenean ut eros et nisl sagittis vestibulum.

Nulla facilisi. Integer lacinia sollicitudin massa. Cras metus. Sed aliquet risus a tortor. Integer id quam.

CHAPTER 1

Introduction

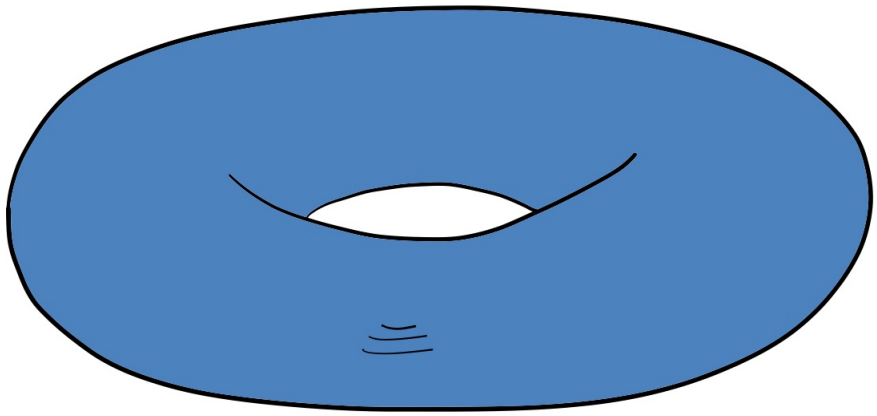

FIGURE 1. This is a torus

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & A' \arrow[r, "f'"] \arrow[d, "a"'] & B' \arrow[r, "g'"] \arrow[d, "b"'] & C' \arrow[r] \arrow[d, "c"'] & 0 \\
0 \arrow[r] & A \arrow[r, "f"] \arrow[d, "a''"'] & B \arrow[r, "g"] \arrow[d, "b''"'] & C \arrow[r] \arrow[d, "c''"'] & 0 \\
0 \arrow[r] & A'' \arrow[r, "f''"] \arrow[d] & B'' \arrow[r, "g''"] \arrow[d] & C'' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

FIGURE 2. The Snake Lemma

This is the introduction chapter. We cite some classic works [**3, 4**].

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Morbi tincidunt nisi sed neque volutpat, et imperdiet metus congue. Donec interdum, neque at varius vehicula, elit odio tempus sapien, sed finibus lorem felis nec turpis. Aliquam erat volutpat. Mauris vehicula imperdiet erat, non aliquet magna commodo id.

Sed porttitor urna ac ipsum rhoncus, vel dictum risus commodo. Integer ultrices dui sit amet pulvinar tempor. Phasellus nec tellus eu turpis sodales accumsan. Proin ultr

THEOREM 1.1. *This is a theorem*

We reference Theorem 1.1.

University of Illinois Chicago.

1. Motivation

1.1. Historical context. A brief overview of how the problem developed.

$$\int_0^1 f(x)dx = 2 \tag{1}$$

How to solve (1)

1.2. Open questions. Some questions remain open for future work.
Note: I have not tested the accessibility of this table.

Monkeys	Lions
100	200

Table 1. Example Table

CHAPTER 2

Background

This chapter gives necessary background.

1. Group theory

A group is a set G with a binary operation satisfying closure, associativity, identity, and inverses.

THEOREM 2.1. *Every finite subgroup of the multiplicative group of a field is cyclic.*

PROOF. This is a standard result from algebra. $\square$

CHAPTER 3

Main Result

1. Richardson varieties

TODO: This maybe belongs in an introduction / background section instead.

For any flag $x \in \mathcal{F}^3$, consider the irreducible Schubert varieties associated with x. We use the notation $x = (l, H)$ where $l \in \mathbb{CP}^2$ and $H \in (\mathbb{CP}^2)^*$ such that $x_1 \subset x_2$.

- $\{x\} = \{(l, H)\}$,
- $\alpha_1(x) = \{(l, H_l) : l \subset H_l\}$,
- $\alpha_2(x) = \{(l_H, H) : l_H \subset H\}$,
- $\alpha_{12}(x) = \{(k, H_l) : l \subset H_l\}$,
- $\alpha_{21}(x) = \{(l_H, G) : l_H \subset H\}$, and
- $\alpha_{121}(x) = \alpha_{212}(x) = \mathcal{F}^3$.

We call a pair of flags $x, y \in \mathcal{F}^3$ *transverse* if they are in general position, meaning that for any proper Schubert variety type α, then $x \notin \alpha(y)$ and $y \notin \alpha(x)$.

Let $x, y \in \mathcal{F}^3$ transverse. Define $R^0_{x,y}$ to be their irreducible 0-dimensional Richardson varieties, which are the intersections between irreducible Schubert varieties $\alpha(x) \cap \beta(y)$ where $dim(\alpha) + dim(\beta) = dim(\mathcal{F}^3)$. There are exactly six such varieties $\{x, x_y, x^y, y_x, y^x, y\}$, where:

- $x := \{x\} \cap \alpha_{121}(y)$,
- $x_y := \alpha_1(x) \cap \alpha_{12}(y)$,
- $x^y := \alpha_2(x) \cap \alpha_{21}(y)$,
- $y_x := \alpha_{12}(x) \cap \alpha_1(y)$,
- $y^x := \alpha_{21}(x) \cap \alpha_2(y)$, and
- $y := \alpha_{121}(x) \cap \{y\}$.

These are explicit points in the flag variety.

Example 3.1. *Suppose x and y are the attracting and repelling flags of $\rho(g) = diag(\lambda, \mu, \nu)$ where $|\lambda| > |\mu| > |\nu|$. Then $x = (e_1, e_1 + e_2)$, $x_y = (e_1, e_1 + e_3)$, $x^y = (e_2, e_1 + e_2)$, $y_x = (e_3, e_1 + e_3)$, $y^x = (e_2, e_2 + e_3)$, and $y = (e_3, e_2 + e_3)$.*

For $x, y \in \mathcal{F}^3$ transverse, define $R^1_{x,y}$ to be irreducible 1-dimensionsal Richardson varieties. Similarly, these are intersections of irreducible Schubert varieties $\alpha(x) \cap \beta(y)$ where $dim(\alpha) + dim(\beta) = dim(\mathcal{F}^3) + 1$. There are six such varieties, each homeomorphic to $\mathbb{CP}^1$, which we distinguish using pairs of elements of $R^0_{x,y}$.

- $span(x, x_y) := \alpha_1(x) \cap \alpha_{121}(y)$,
- $span(x, x^y) := \alpha_2(x) \cap \alpha_{121}(y)$,
- $span(x_y, y_x) := \alpha_{12}(x) \cap \alpha_{12}(y)$,
- $span(x^y, y^x) := \alpha_{21}(x) \cap \alpha_{21}(y)$,
- $span(y_x, y) := \alpha_{121}(x) \cap \alpha_1(y)$, and
- $span(y^x, y) := \alpha_{121}(x) \cap \alpha_2(y)$.

These varieties are not pairwise disjoint, e.g. $\{x\} = span(x, x_y) \cap span(x, x^y)$. The points of intersection between 1-dimensional Richardson varieties are the 0-dimensional Richardson varieties.

Remark 3.1. *If $x, y \in \mathcal{F}^3$ transverse and $dim(\alpha) + dim(\beta) < dim(\mathcal{F}^3)$, then $\alpha(x) \cap \beta(y) = .$*

2. Assumptions and notation

Let Γ the fundamental group of a closed hyperbolic surface. Let $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$ Anosov. We use the following notation related to ρ and ρ'.

- $\xi, \xi' : \partial\Gamma \to \mathcal{F}^3$ refers to the continuous, transverse, ρ- or ρ'- equivariant, dynamics-preserving boundary maps.
- $\Lambda, \Lambda' \subset \mathcal{F}^3$ refers to the limit curves, which are the images of ξ, ξ'.
- $Th(x) = \alpha_1(x) \cup \alpha_2(x)$ refers to the balanced ideal thickening of $x \in \mathcal{F}^3$.
- $Th(\Lambda) = \cup_{x\in\Lambda} Th(x)$ refers to the thickening of the limit curve.

We call ρ, ρ' *topologically conjugate* if there exists a homeomorphism $F : \mathcal{F}^3 \to \mathcal{F}^3$ such that for all $g \in \Gamma$ and all $x \in \mathcal{F}^3$, then $F(\rho(g) \cdot x) = \rho'(g) \cdot F(x)$. We call F a *topological conjugacy.*

TODO: definition of hyperconvexity

3. Statement of Theorem 3.1 and related results

THEOREM 3.1. *Hyperconvexity is closed under topological conjugacy. In other words, if ρ and ρ' are topologically conjugate, then ρ is hyperconvex if and only if ρ' is hyperconvex.*

We use two supporting lemmas to prove this theorem. First, we show that the relative position of 0-dimensional Richardson varieties of Λ determines whether ρ is hyperconvex.

LEMMA 3.1.1. *An Anosov representation ρ is hyperconvex* if and only if *for all $x, y, z \in \Lambda^{(3)}$ we have $R^0_{x,y} \cap R^0_{x,z} = \{x\}$.*

For the second lemma, assume that $F : \mathcal{F}^3 \to \mathcal{F}^3$ is a topological conjugacy between ρ and ρ'. Clearly $F(\Lambda) = \Lambda'$. This mapping is pointwise, i.e. for all $\alpha \in \partial\Gamma$, then $F(\xi(\alpha)) = \xi'(\alpha)$. So for $x \in \Lambda$, we can use the notation $x' := F(x) \in \Lambda'$.

Topological conjugacies may preserve the 0- and 1-dimensional Richardson varieties of Λ in one of two ways. We say F *matches* the Richardson varieties if for all $x, y \in \Lambda^{(2)}$, then $F(x_y) = x'_{y'}$ and $F(span(x, x_y)) = span(x, x'_{y'})$. We say F *switches* the Richardson varieties if for all $x, y \in \Lambda^{(2)}$, then $F(x_y) = x'^{y'}$ and $F(span(x, x_y)) = span(x, x'^{y'})$.

LEMMA 3.1.2. *For all $x, y \in \Lambda^{(2)}$, then $F(R^0_{x,y}) = R^0_{x',y'}$ and $F(R^1_{x,y}) = R^1_{x',y'}$. Furthermore, F either matches or switches the Richardson varieties.*

We also have the following consequence of Lemma 3.1.2, which is of independent interest:

COROLLARY 3.1.1. *For all $x \in \Lambda$, then $F(Th(x)) = Th(x')$.*

4. Proof of Theorem 3.1.

We first prove our main theorem using our supporting lemmas.

PROOF. (Theorem 3.1) Suppose $F : \mathcal{F}^3 \to \mathcal{F}^3$ is a topological conjugacy between ρ and ρ'.

We begin with the forward direction. Assume that ρ is hyperconvex. Let $x, y, z \in \Lambda^{(3)}$. By Lemma 3.1.1, then $x_y \neq x_z$ and $x^y \neq x^z$. By Lemma 3.1.2, then the sets

$$\{F(x_y), F(x^y), F(x_z), F(x^z)\} = \{x'_{y'}, x'^{y'}, x'_{z'}, x'^{z'}\} \tag{2}$$

contain four distinct points. Since x, y, z were general, then this holds for all triples in Λ, and ρ' is hyperconvex.

The other direction proceeds similarly. Assume that ρ is *not* hyperconvex. By Lemma 3.1.1, there exist $x, y, z \in \Lambda^{(3)}$ such that $x_y = x_z$ or $x^y = x^z$. By Lemma 3.1.2, then F either matches or switches Richardson varieties. So one of the following is true: $x'_{y'} = x'_{z'}$ or $x'^{y'} = x'^{z'}$. Therefore ρ' is *not* hyperconvex. □

We also prove our lemma linking Richardson 0-dimensional varieties to hyperconvexity.

Proof. (Lemma 3.1.1) Suppose ρ is Anosov. Let $x = (l_x, H_x), y = (l_y, H_y), z = (l_z, H_z) \in \Lambda^{(3)}$. Since x, y, z are pairwise transverse, then

$$y_x, y^x, z_x, z^x \notin R^0_{x,y} \cap R^0_{x,z}. \tag{3}$$

It suffices to check the pairs $x_y, x_z \in \alpha_1(x)$ and $x^y, x^z \in \alpha_2(x)$.

By definition, $x_y = (l_x, l_x + l_y)$ and $x_z = (l_x, l_x + l_z)$, which are distinct if and only if $l_x + l_y \neq l_x + l_z$, or equivalently if and only if $l_z \not\subset l_x + l_y$.

Similarly, $x^y = (H_x \cap H_y, H_x)$ and $x^z = (H_x \cap H_z, H_x)$, which are distinct if and only if $H_x \cap H_y \neq H_x \cap H_z$, or equivalently if and only if $H_x \cap H_y \not\subset H_z$.

So $R^0_{x,y} \cap R^0_{x,z} = \{x\}$ for all $x, y, z \in \Lambda^{(3)}$ is equivalent to ρ hyperconvex. □

5. Action of $\rho(g)$ on F^3

The remainder of this chapter is devoted to the proof of Lemma 3.1.2.

In this section we describe the action of $\rho(g)$ on $\mathcal{F}^3$ for $g \in \Gamma$ infinite order. Since ρ is Anosov, then $\rho(g)$ is strongly loxodromic, i.e. its eigenvalues have distinct magnitude. That means $\rho(g)$ has attracting and repelling limit curve points $x = \xi(g^+)$ and $y = \xi(g^-)$, respectively.

Under the action $\rho(g)$, most of $\mathcal{F}^3$ is attracted to x and repelled by y. We define the *uphill* of x as the subset attracted to x under $\rho(g)$:

$$U_{\rho(g)}(x) := \{f \in F^3 : \rho(g^n) \cdot f \to x\}. \tag{4}$$

Similarly, we define the *downhill* of y as the subset repelled by y under $\rho(g)$, or equivalently the subset attracted by y under $\rho(g^{-1})$:

$$D_{\rho(g)}(y) = U_{rho(g^{-1})}(y) := \{f \in \mathcal{F}^3 : \rho(g^{-n}) \cdot f \to y\}. \tag{5}$$

We then define the *exceptional set* of $\rho(g)$ as the subset *not* attracted to x or *not* repelled by y under $\rho(g)$:

$$E_{\rho(g)} = \mathcal{F}^3 \setminus (U_{\rho(g)}(x) \cap D_{\rho(g)}(y)). \tag{6}$$

The set $E_{\rho(g)}$ is nonempty and closed, since it is complement of an open set. It contains the fixed points of $\rho(g) \cdot \mathcal{F}^3$, which are precisely $R^0_{x,y}$. It suffices to show this for $x, y, \rho(g)$ as in Example 3.1, since up to conjugation $\rho(g) = diag(\lambda, \mu, \nu)$.

We can partition $E_{\rho(g)}$ using each element's attracting and repelling fixed points. By taking the closure of these subsets, we recover the 1-dimensional Richardson varieties $R^1_{x,y}$.

- $span(x, x_y) = \alpha_1(x) = \overline{U(x) \cap D(x_y)}$
- $span(x, x^y) = \alpha_2(x) = \overline{U(x) \cap D(x^y)}$
- $span(x_y, y_x) = \overline{U(x_y) \cap D(y_x)}$
- $span(x^y, y^x) = \overline{U(x^y) \cap D(y^x)}$
- $span(y_x, y) = \alpha_1(y) = \overline{U(y_x) \cap D(y)}$
- $span(y^x, y) = \alpha_2(y) = \overline{U(y^x) \cap D(y)}$

So $E_{\rho(g)}$ is homeomorphic to the union of six copies of $\mathbb{CP}^1$, each identified with two neighbors in a cyclic order $(x, x_y, y_x, y, y^x, x^y)$.

6. F preserves Richardson 0- and 1-varieties of $\xi(g^+), \xi(g^-)$

Assume $F : \mathcal{F}^3 \to \mathcal{F}^3$ is a topological conjugacy between $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$. We use the notation $x' = F(x), y' = F(y)$.

We first consider the attracting and repelling flags for an infinite order element $g \in \Gamma$.

Proposition 3.1. *Let $x, y \in \Lambda$ such that $x = \xi(g^+)$ and $y = \xi(g^-)$. Then all of the following are true:*

(1) $F(R^0_{x,y}) = R^0_{x',y'}$,
(2) $F(R^1_{x,y}) = R^1_{x',y'}$, *and*
(3) F *either matches or switches the Richardson varieties of x and y.*

Proof. Since F is a topological conjugacy, it must preserve the exceptional set of $\rho(g)$, so $F(E_{\rho(g)}) = E_{\rho'(g)}$. Similarly, since F preserves the limit curve pointwise, then $F(x) = x'$ and $F(y) = y'$.

Since $F|_{E_{\rho(g)}}$ is a homeomorphism, it sends 1-varieties to 1-varieties, and sends intersections of elements of $R^1_{x,y}$ to intersections of elements of $R^1_{x',y'}$. This proves (1) and (2).

To prove (3), we observe that:

$$F(x) = F(span(x, x_y) \cap span(x, x^y)) \tag{7}$$
$$= F(span(x, x_y)) \cap F(span(x, x^y)) \tag{8}$$
$$= span(x', x'_{y'}) \cap span(x', x'^{y'}) \tag{9}$$
$$= x'. \tag{10}$$

So we conclude that either $F(x_y) = x'_{y'}$ and $F(span(x, x_y)) = span(x', x'_{y'})$, or $F(x_y) = x'^{y'}$ and $F(span(x, x_y)) = span(x', x'^{y'})$. □

7. F preserves Richardson 0-varieties of Λ

In this section, we still assume $F : \mathcal{F}^3 \to \mathcal{F}^3$ is a topological conjugacy between ρ, ρ'. We extend our argument from the Richardson 0-varieties of pairs $\xi(g^+), \xi(g^-)$ to the Richardson 0-varieties of all distinct pairs in Λ.

First, we extend Proposition 3.1 to Richardson varieties of pairs x, z where x is the attracting fixed point of an infinite order element $g \in \Gamma$ and z is a (distinct) generic point on Λ.

Proposition 3.2. *Let $g \in \Gamma$ infinite order, and define $x := \xi(g^+)$. Let $z \in \Lambda \setminus \{x\}$. Then $F(x_z) \in \{x'_{z'}, x'^{z'}\}$.*

We use the following result from group theory:

Remark 3.2 (TODO REFERENCE). *Let Γ hyperbolic with boundary $\partial\Gamma \equiv S^1$, and let $x, z \in \partial\Gamma$. Then there exists a quasigeodesic sequence $\{g_i\}_{i=1} \in \Gamma^{\mathbb{N}}$ such that for all $y \in \partial\Gamma$:*

$$lim_{i\to\infty} g_i \cdot y = \begin{cases} x : y = x \\ z : y \neq x \end{cases} . \tag{11, 12}$$

PROOF. (Proposition 3.2) If $z = \xi(g^-)$ then we are done by Prop 3.1, so assume not. Set $y := \xi(g^-)$.

Let $\{g_i\}_{i=1} \in \Gamma^{\mathbb{N}}$ as in Rem. 3.2, such that $\lim_{i=1} \rho(g_i) \cdot y = z$ and $lim_{i=1}\rho(g_i) \cdot x = x$. Let $x_i := \rho(g_i) \cdot x$ and $y_i := \rho(g_i) \cdot y$, and let $x'_i := F(x_i), y'_i := F(y_i)$.

By Prop. 3.1 then for all $i \in \mathbb{N}$ we have $F(x_{iy_i}) \in \{x'_{iy'_i}, x'^{y'_i}_i\}$. Clearly $\lim_{i\to\infty} x_{iy_i} = x_z$, $\lim_{i\to\infty} x'_i = x'$ and $\lim_{i\to\infty} y'_i = z'$. By continuity of F and compactness of $\mathcal{F}^3$, then

$$F(x_z) = F(\lim_{i\to\infty} x_{iy_i}) \tag{13}$$
$$= \lim_{i\to\infty} F(x_{iy_i}) \tag{14}$$
$$\in \{\lim_{i\to\infty} x'_{iy'_i}, \lim_{i\to\infty} x'^{y'_i}_i\} \tag{15}$$
$$= \{x'_{z'}, x'^{z'}\}. \tag{16}$$

□

Next, we extend Proposition 3.2 to Richardson varieties of the form w_z where both w and z are generic points of Λ.

PROPOSITION 3.3. *For all $w, z \in \Lambda^{(2)}$, then $F(w_z) \in \{w'_{z'}, w'^{z'}\}$.*

PROOF. If w is the attracting fixed point of an infinite order element, we are done by Prop 3.2, so assume not.

Let $\{x_i\}_{i=1} \in \Lambda^{\mathbb{N}}$ such that each $x_i = \xi(g_i)$ the attracting fixed point of an infinite order word, and $\lim_{i\to\infty} x_i = w$. This sequence exists because attracting fixed points of $\rho(g)$ (for $g \in \Gamma$ infinite order) are dense in Λ. Define $x'_i = F(x_i)$. By continuity of F, we have

$$F(w) = F(\lim_{i\to\infty} x_i) \tag{17}$$
$$= \lim_{i\to\infty} F(x_i) \tag{18}$$
$$= \lim_{i\to\infty} (x'_i) \tag{19}$$
$$= w'. \tag{20}$$

Then by Prop 3.2 and by continuity of F, we have

$$F(w_z) = F(\lim_{i\to\infty} x_{iz}) \tag{21}$$
$$= \lim_{i\to\infty} F(x_{iz}) \tag{22}$$
$$\in \{\lim_{i\to\infty} (x'_{iz'}), \lim_{i\to\infty} (x'^{z'}_i)\} \tag{23}$$
$$= \{w'_{z'}, w'^{z'}\}. \tag{24}$$

□

8. F preserves Richardson 1-varieties of Λ

In this section, we highlight how F preserves the Richardson 1-varieties of Λ, in particular the balanced ideal thickening centered at each point $x \in \Lambda$.

Observe that $Th(x) = \alpha_1(x) \cup \alpha_2(x) = span(x, x_y) \cup span(x, x^y)$ for some $y \in \Lambda \setminus \{x\}$.

Proof. (Corollary 3.1.1) By Prop. 3.1, $\{F(x_y), F(x), F(x^y)\} = \{x'_{y'}, x, x'^{y'}\}$. Also, Observe that F is a homeomorphism that preserves dynamics in that $F(U_{\rho(g)}) = U_{\rho'(g)}$ and $F(D_{\rho(g)}) = D_{\rho'(g)}$. So then

$$F(Th(x)) = F(span(x, x_y) \cup span(x, x^y)) \tag{25}$$
$$= span(x', x'_{y'}) \cup span(x', x'^{y'}) \tag{26}$$
$$= Th(x'). \tag{27}$$

□

If F *matches* Richardson varieties, then for all $x \in \Lambda$ we have $F(x_y) = x'_{y'}$ and $F(\alpha_1(x)) = \alpha_1(x')$. Otherwise, if F *switches* Richardson varieties, then for all $x \in \Lambda$ we have $F(x_y) = x'^{y'}$ and $F(\alpha_1(x)) = \alpha_2(x')$.

Corollary 3.1.2. *F either matches of switches Richardson varieties.*

Proof. Suppose for some $w \in \Lambda$ we have $F(\alpha_1(w)) = \alpha_1(w')$. Let $z \in \Lambda\ \{w\}$. Since F is a homeomorphism, by Cor. 3.1.1 we have $F(\alpha_1(z)) \in \{\alpha_1(z'), \alpha_2(z')\}$.

For contradiction, suppose that $F(\alpha_1(z)) = \alpha_2(z')$. Then for all $g \in \Gamma$, by the definition of a topological conjugacy, we have $F(\alpha_1(\rho(g) \cdot z)) = \alpha_2(\rho'(g) \cdot z')$.

Since $\rho(g) \cdot z$ is dense in Λ, then there exists a sequence $\{g_i\}_{i=1} \in \Gamma^{\mathbb{N}}$ such that $\lim_{i\to\infty} \rho(g_i) \cdot z = w$. Define $z_i = \rho(g_i) \cdot z$.

Then

$$\alpha_1(w') = F(\alpha_1(w)) \tag{28}$$
$$= F(\alpha_1(\lim_{i\to\infty} z_i)) \tag{29}$$
$$= \lim_{i\to\infty} F(\alpha_1(z_i)) \tag{30}$$
$$= \lim_{i\to\infty} \alpha_2(w) \tag{31}$$
$$= \alpha_2(w). \tag{32}$$

This is a contradiction. □

Clearly $x_y \in \alpha_1(x)$, so if F matches Richardson varieties then $F(x_y) = x'_{y'}$, and if F switches Richardson varities then $F(x_y) = x'^{y'}$.

So for all $x, y \in \Lambda$, a topological conjugacy F sends $R^0_{x,y}$ to $R^0_{x',y'}$, and either matches or switches the irreducible 0- and 1-dimensional Richardson varieties. This completes the proof of Lemma 3.1.2.

9. Comments

Remark 3.3. *For $x, y \in (\mathcal{F}^3)^{(2)}$, F will not necessarily preserve the 0-dimensional Richardson varieties.*

Remark 3.4. *Theorem 3.1 applies to virtual hyperbolic surface groups as well, e.g. $\Gamma = \pi_1(\mathcal{O}^2)$ where $\mathcal{O}^2$ is a hyperbolic 2-orbifold.*

TODO: What about higher dimensions?

CHAPTER 4

Another result

1. Notation

2. Statement of Theorem 4.1 and related results

THEOREM 4.1. *Suppose $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$ in the same Anosov component. Suppose furthermore that ρ and ρ' are irreducible. Then ρ, ρ' are topologically conjugate.*

Combined with Theorem 3.1, we get the following corollaries:

COROLLARY 4.1.1. *All quasi-Hitchin representations are hyperconvex. All quasi-Barbot representations are* not *hyperconvex.*

Furthermore:

COROLLARY 4.1.2. *The quasi-Hitchin and quasi-Barbot Anosov components are distinct.*

We also consider the topological conjugacies of reducible representations $\mathcal{A}_{red}$.

THEOREM 4.2. *There are exactly three topological conjugacy classes of $\mathcal{A}_{red}(\Gamma)$:*

- $[\rho_{ss}]$
- $[\rho_{line}] = [\rho_{plane}]$
- $[\rho_{dense}]$

3. Key ingredients

In this section, we highlight some key results that will be used in the proof of Theorem 4.1. First, we use the following theorem of Stecker:

THEOREM 4.3 (**??**, Thm 3.24). *Let $\rho : \Gamma \to G$ Borel Anosov. If $dim(\Lambda) \leq mbic(G) - 2$, then every non-empty cocompact domain of discontinuity in $\mathcal{F}^3$ is dense and connected, and there is a bijection:*

$$\{\textit{balanced ideals in } W(G)\} \leftrightarrow \{\textit{nonempty cocompact domains of discontinuity in } \mathcal{F}(G)\} \tag{33}$$

We tailor Stecker's theorem to our assumptions. Recall that $G = SL_3\mathbb{C}$, so $mbic(G) = 4$ and $\mathcal{F}(G) = \mathcal{F}^3$. Then $W(G) = A_2$ which has exactly one balanced ideal. In $SL_3\mathbb{C}$, all Anosov representations are Borel Anosov. Furthermore, we assume that Γ is a hyperbolic surface group, so $\partial\Gamma \cong S^1$. Then $dim(\Lambda_\rho) \leq mbic(G) - 2 = 2$ and the condition in Theorem 4.3 is met.

REMARK 4.1. *For Γ a hyperbolic surface group and $\rho : \Gamma \to SL_3\mathbb{C}$ Anosov, then Ω_ρ is the unique nonempty cocompact domain of discontinuity. Furthermore, $\Omega_r ho$ is connected and dense in $\mathcal{F}^3$.*

We also use the following lemma, proved in section

LEMMA 4.3.1. *For Γ a hyperbolic surface group and $\rho : \Gamma \to SL_3\mathbb{C}$ Anosov, then Ω_ρ is simply connected.*

Finally, we use the following theorem of Dumas and Sanders:

THEOREM 4.4 (**??**, Thm 5.1). *Let (P, Q) be parabolic subgroups of G, with P symmetric, and let $I \subset W(G)$ be a balanced ideal of type (P, Q). Let $\rho : \Gamma \to G$ be a P-Anosov representation. Then for any $\rho' \in \mathcal{A}(\rho, P)$, the quotient manifolds M^I_ρ and $M^I_{\rho'}$ are diffeomorphic.*

Again, we tailor this theorem to our assumptions that $G = SL_3\mathbb{C}$, where there is only one balanced ideal and one .

REMARK 4.2. *For Γ a hyperbolic surface group and $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$ in the same Anosov component, then*

$$M_\rho \equiv_{diffeo} M_{\rho'}. \tag{34}$$

4. Proof of Theorem 4.1

PROOF. (Theorem 4.1) Let $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$ irreducible and in the same Anosov component. We want to show the existence of $F : \mathcal{F}^3 \to \mathcal{F}^3$ a topological connjugacy of ρ, ρ'.

Then by Thm. 4.4, there exists a diffeomorphism $D : M_\rho \to M_{\rho'}$. This lifts to a diffeomorphism of the universal cover $\tilde{D} : \tilde{M}_\rho \to \tilde{M}_{\rho'}$. Then by Lemma 4.3.1, the domains of discontinuity Ω_ρ, $\Omega_{\rho'}$ are simply connected, so $\tilde{M}_\rho \equiv \Omega_\rho$ and $\tilde{M}_{\rho'} \equiv \Omega_{\rho'}$. So we can consider $\tilde{D} : \Omega_\rho \to \Omega_{\rho'}$.

Clearly $\tilde{D}$ is a topological conjugacy of Ω_ρ and $\Omega_{\rho'}$, since it is a diffeomorphism and for all $f \in \Omega_\rho$ and $g \in \Gamma$ we have $\tilde{D}(\rho(g) \cdot f) = \rho'(g) \cdot \tilde{D}(f)$.

Next, we want to extend $\tilde{D}$ to a homeomorphism $F : \mathcal{F}^3 \to \mathcal{F}^3$ such that $F|_{\Omega_\rho} \equiv \tilde{D}$. By Theorem 4.3, Ω_ρ is an open, connected dense subset of $\mathcal{F}^3$, so $\tilde{D}$ has a unique continuous extension to $\mathcal{F}^3$ using sequences in Ω_ρ which limit to points in $Th(\Gamma)$.

TODO check thet F is a topological conjugacy. TODO where do we use irreducible? □

REMARK 4.3. *The topological conjugacy F may not be unique, however it is uniquely determined by the choice of the diffeomorphism D and its lift $\tilde{D}$.*

5. Topological conjugacies of reducible representations

This section contains the proof of Theorem 4.2

CHAPTER 5

Vitae

For UIC this can be a full CV or a short version (e.g. half a page) with previoius degrees, work experience and publications.

APPENDIX A

Technical Lemmas

Here we collect some supporting lemmas.

Bibliography

[1] E. Dumas, and A.Sanders "Geometry of compact complex manifolds associated to generalized quasi-Fuchsian varieties" (2017)

[2] F. Stecker, "Balanced ideals and domains of discontinuity of Anosov representations" (2022)

[3] R. Hartshorne, *Algebraic Geometry*, Springer-Verlag, New York, 1977.

[4] D. Mumford, *Abelian Varieties*, Oxford University Press, 1970.

[5] J. Draisma, E. Horobet, G. Ottaviani, B. Sturmfels, and R. R. Thomas, "The Euclidean distance degree of an algebraic variety," *arXiv:1309.0049* (2013). Available at: https://arxiv.org/abs/1309.0049